THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these is a rational number? 1

(a) $\sqrt{5}+\sqrt{6}$ **(b)** $\dfrac{\sqrt{75}}{\sqrt{3}}$ **(c)** $5\sqrt{8}$ **(d)** $\sqrt{7}$

2. A number ends with exactly 3 zeros if its prime factorisation (other factor 3) is: 1

(a) $3^3 \times 7$ **(b)** $2^4 \times 5^2 \times 3$ **(c)** $2^4 \times 5^3 \times 3$ **(d)** $2^2 \times 5^3 \times 3$

3. The LCM of 45, 60 and 75 is: 1

(a) 2700 **(b)** 450 **(c)** 900 **(d)** 6750

4. The least number that is divisible by every number from 1 to 8 is: 1

(a) 840 **(b)** 420 **(c)** 1680 **(d)** 20160

5. The largest number dividing 85, 145 and 205 leaving remainder 5 in each case is: 1

(a) 20 **(b)** 40 **(c)** 30 **(d)** 10

6. The smallest positive integer by which 45 must be multiplied to obtain a perfect square is: 1

(a) 5 **(b)** 15 **(c)** 45 **(d)** 3

7. Which of these is a rational number? 1

(a) $6\sqrt{9}$ **(b)** $\sqrt{6}+\sqrt{7}$ **(c)** $\dfrac{\sqrt{108}}{\sqrt{3}}$ **(d)** $\sqrt{8}$

8. A number ends with exactly 4 zeros if its prime factorisation (other factor 3) is: 1

(a) $2^3 \times 5^4 \times 3$ **(b)** $2^5 \times 5^3 \times 3$ **(c)** $3^4 \times 7$ **(d)** $2^5 \times 5^4 \times 3$

9. **Assertion (A):** $\sqrt{2} \times \sqrt{8}$ is a rational number. 1

Reason (R): The product of two irrational numbers is always irrational.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The decimal expansion of $\frac{7}{80}$ is terminating. 1

Reason (R): The denominator $80 = 2^4 \times 5$ has no prime factor other than 2 and 5.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the smallest positive integer that leaves remainders 11 and 16 when divided by 15 and 20 respectively. 2

12. Prove that $5\sqrt{2}$ is irrational, given that $\sqrt{2}$ is irrational. 2

13. Two numbers have HCF 9 and are in the ratio $5 : 8$. Find the numbers and their LCM. 2

14. Find the smallest positive integer that leaves remainders 4 and 8 when divided by 10 and 14 respectively. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A, B and C take 15, 40 and 60 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

16. Find the greatest number that divides 2013 and 2517 leaving remainder 3 in each case. 3

17. Is $7 \times 8 \times 9 + 9$ a composite number? Justify. 3

OR

A, B and C take 16, 24 and 30 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

18. Find the greatest number that divides 1507 and 1927 leaving remainder 7 in each case. 3

OR

Is $8 \times 9 \times 10 + 10$ a composite number? Justify. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two runners start together from the same point of a circular track. Aman takes 36 seconds and Bhavna takes 48 seconds to complete one round, and both keep a constant pace. 5

(i) Write 36 and 48 in prime factorised form.

(ii) After how many seconds will they meet again at the starting point?

(iii) How many rounds will each have completed by then?

(iv) A third runner, Chetan, takes 60 seconds per round. After how many seconds will all three meet again at the starting point, and how many rounds will Chetan have run?

20. Three sets of English, Hindi and Mathematics books have 168, 252 and 420 books. They are stacked so that each stack has the same number of books of one subject and all stacks have the same height. Find the minimum number of stacks. 5

OR

Prove that $\frac{1}{\sqrt{7}}$ is irrational and hence that $\frac{4}{\sqrt{7}}$ is irrational. 5

END OF QUESTION PAPER — **Total: 40 Marks**